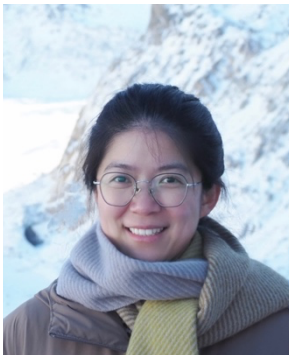
Dr. Hailing Jia



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Professional career

2023.09 – Present	Tenure-track Scientist Space Research Organization Netherlands (SRON)
2020.10 – 2023.08	Postdoctoral Researcher Clouds and Global Climate working group (Prof. Johannes Quaas) Institute for Meteorology, University of Leipzig
2018.12 – 2019.12	Visiting scholar Atmospheric Sciences Research Center State University of New York at Albany

Education

2015.09 – 2020.06	Ph.D., Atmospheric Physics and Environment Science, Nanjing University of Information Science and Technology, China
2011.09 – 2015.06	B.S., Atmospheric Physics, Nanjing University of Information Science and Technology, China

Research Focus

- Aerosol-cloud-radiation interactions
- Cloud processes and cloud-climate feedback
- Climate modeling
- In situ and satellite observations

National Funding

- 2023.09 – 2028.09 NWO (Dutch Research Council) WISE Fellowship
- 2025.09 – 2029.09 NWO ENW-M1 grant (**PI**): *Tackling Challenges in Aerosol–Cloud InterACTIONs with a Novel Framework and New Satellites (ACIaction, €0.34 M)*

Publications

2026

1. **Jia H.*,** Quaas J., Kroese W., Diedenhoven B., Gryspeerdt E., Bohm C., Block K., and Hasekamp O. Optimal choice of proxy for cloud condensation nuclei reduces uncertainty in aerosol–cloud–climate forcing. ***Science Advances***, 12(8), 2026. https://doi.org/10.1126/sciadv.aea4828

2. Bhatti, Y. A., Watson-Parris, D., Regayre, L. A., **Jia, H**., Neubauer, D., Im, U., Svenhag, C., Schutgens, N., Tsikerdekis, A., Nenes, A., Irfan, M., van Diedenhoven, B., Arifi, A., Fu, G., and Hasekamp, O. P.: Uncertainty in aerosol effective radiative forcing from anthropogenic and natural aerosol parameters in ECHAM6.3-HAM2.3, ***Atmos. Chem. Phys.,*** 26, 269–293, https://doi.org/10.5194/acp-26-269-2026, 2026.

3. Hannadige N., Fu G., Diedenhoven B., **Jia H.,** and Hasekamp O. Estimation of Cloud Condensation Nuclei (CCN) from SPEXone on PACE using a deep neural network retrieval algorithm. ***J. Quant. Spectrosc. Radiat. Transfer.*** (in revision).

4. Sun Y., Zhao C., **Jia H.** and Yang Y. Absorbing aerosols exacerbating the heterogeneity of precipitation profiles. ***npj Clim. Atmos. Sci.*** (in revision)

5. Wang H., Peng Y., **Jia H**., Huang J., Zhao Y., Shi C., Shang H., Baran A., Yin S., Riedi J., Quaas J. Constraining Aerosol–Cloud Interactions via the Droplet–Aerosol Trends Discrepancy. ***Nature Geoscience*** (submitted)

6. Wang Y., Fang F., Li J., Zhang P., Zhao J., **Jia H.,** Huang J. Overestimation of the Twomey Effect in Orographic Clouds over the Southern Tibetan Plateau Due to Neglected Dynamical Constraints. ***Geophys. Res. Lett.*** (in review).

7. Zhu Y., Yang Y., **Jia H.,** Wang H., Wang P. and Liao H. Strategizing Emission Cuts to Reduce the Unintended Warming Consequence of Clean Air. ***Nat. Commun.*** (in review)

8. CLÉMENT N., Bellouin N., Boucher O., Quaas J., **Jia H.** and Tegen I. Aerosol and Non-Aerosol Drivers of Regional Trends in Top-of-Atmosphere Albedo over 2002–2020. ***Geophys. Res. Lett.*** (in review).

9. Liu Y., **Jia H**., and Han, Y.: Susceptibility of Marine Warm Clouds to Aerosols in Different Monsoon Periods over the South China Sea, EGUsphere [preprint], https://doi.org/10.5194/egusphere-2025-5935, 2026.

2025

10. Henkes A., **Jia H.,** Kretzschmar J., Hörnig S., and Quaas J. Hemispheric contrast in aerosol-cloud interactions: an attempt for detection and attribution. ***Tellus B: Chem. phys. Meteorol.***, 77(1), p. 41–58, 2025.

2024

11. Hodnebrog Ø., Myhre G., Jouan C., Andrews T., Forster P., **Jia H**., Loeb N., Olivié D., Paynter D., Quaas J., Raghuraman S. and Schulz M. Recent reductions in aerosol emissions have increased Earth's energy imbalance. ***Commun. Earth Environ.,*** 5, 166, 2024.

12. **Jia H.*,** Hasekamp O. and Quaas J. Revisiting aerosol–cloud interactions from weekly cycles, ***Geophys. Res. Lett.***, 51, e2024GL108266. https://doi.org/10.1029/2024GL108266, 2024.
13. Quaas J., Andrews T., Bellouin N., Block K., Boucher O., Ceppi P., Dagan G., Doktorowski S., Eichholz M., Forster P., Goren T., Gryspeerdt E., Hodnebrog O., **Jia H.**, Kramer R., Lange C., Maycock A., M Mülmenst¨adt J., Myhre G., Pincus R., Samset B., Senf F., Shine K. and Wall C. Adjustments to climate perturbations—mechanisms, implications, observational constraints. ***AGU Adv.***, 5, e2023AV001144. https://doi.org/10.1029/2023AV001144, 2024.
14. Wang Y., **Jia H.**, Zhang P., Fang, F., Li, J., Zhu, L., Wang, Y., Wang, T., Li, J. Sensitivity of cloud microphysics to aerosol is highly associated with cloud water content: Implications for indirect radiative forcing, ***Atmos. Res.***,, 309, 107552, https://doi.org/10.1016/j.atmosres.2024.107552, 2024.
15. Wang Y., Li J., Fang F., Zhang P., He J., Pöhlker M., Henning S., Tang C., **Jia H**., Wang Y., Jian B., Shi J., Huang J. In-situ observations reveal weak hygroscopicity in the Southern Tibetan Plateau: implications for aerosol activation and indirect effects, ***npj Clim. Atmos. Sci.,*** 7, 77, 2024.
16. Zhao, J., Ma, X., Quaas, J., and **Jia, H**.: Exploring aerosol–cloud interactions in liquid-phase clouds over eastern China and its adjacent ocean using the WRF-Chem–SBM model, ***Atmos. Chem. Phys***., 24, 9101–9118, https://doi.org/10.5194/acp-24-9101-2024, 2024.

2023

17. **Jia H.*** and Quaas J. Non-linearity of the cloud response postpones climate penalty of mitigating air pollution in polluted regions, ***Nat. Clim. Change***, **13**, 943–950, 2023.
18. Rosenfeld D., Kokhanovsky A., Goren T., Gryspeerdt E., Hasekamp O., **Jia H.**, Lopatin A., Quaas J., Sourdeval O., and Pan Z. Frontiers in satellite-based estimates of cloud-mediated aerosol forcing, ***Reviews of Geophysics***, 61, e2022RG000799, 2023.
19. Stjern, C. W., P. M. Forster, **H. Jia**, C. Jouan, M. R. Kasoar, G. Myhre, D. Olivié, J. Quaas, B. H. Samset, M. Sand, T. Takemura, A. Voulgarakis, and C. D. Wells: The timescales of climate responses to carbon dioxide and aerosols. ***J. Clim.***, 36, 3537–3551, 2023.

2022

20. Goren, T., Feingold, G., Gryspeerdt, E., Kazil, J., Kretzschmar, J., **Jia, H.**, & Quaas, J. Projecting stratocumulus transitions on the albedo—cloud fraction relationship reveals linearity of albedo to droplet concentrations. ***Geophys. Res. Lett.***, e2022GL101169, 2022.
21. **Jia H.***, Quaas J., Gryspeerdt, E., Böhm, C., and Sourdeval, O.: Addressing the difficulties in quantifying droplet number response to aerosol from satellite observations. ***Atmos. Chem. Phys***., 22, 7353–7372, 2022.
22. Quaas, J., **Jia, H.**, Smith, C., Albright, A. L., Aas, W., Bellouin, N., Boucher, O., Doutriaux-Boucher, M., Forster, P. M., Grosvenor, D., Jenkins, S., Klimont, Z., Loeb, N. G., Ma, X., Naik, V., Paulot, F., Stier, P., Wild, M., Myhre, G., and Schulz, M.: Robust evidence for reversal of the trend in aerosol effective climate forcing, ***Atmos. Chem. Phys***., 22, 12221–12239, 2022.

2021

23. **Jia H.**, Ma X., Yu F., Quaas J.: Significant underestimation of radiative forcing by aerosol–cloud interactions derived from satellite-based methods. ***Nat. Commun.***12, 3649, 2021.

2016 – 2020

24. Chaibou A. A. S., Ma X., Kumar K. R., **Jia H.**, Tang Y., & Sha T.: Evaluation of dust extinction and vertical profiles simulated by WRF-Chem with CALIPSO and AERONET over North Africa. ***J. Atmos. Sol.-Terr. Phys.,***, 199:105213, 2020.
25. **Jia H.**, Ma X., Yu F., Liu Y., and Yin Y.: Distinct Impacts of Increased Aerosols on Cloud Droplet Number Concentration of Stratus/Stratocumulus and Cumulus, ***Geophys. Res. Lett.***, 46, 13517–13525, 2019.

26. **Jia H.**, Ma X., Quaas J., Yin Y., and Qiu T.: Is positive correlation between cloud droplet effective radius and aerosol optical depth over land due to retrieval artifacts or real physical processes?, ***Atmos. Chem. Phys***., 19, 8879–8896, 2019.
27. **Jia H.**, Ma X., and Liu Y.: Exploring aerosol–cloud interaction using VOCALS-REx aircraft measurements, ***Atmos. Chem. Phys***., 19, 7955–7971, 2019.
28. **Jia H**., Ma X., and Xiong F.: Spatial and temporal distributions of warm cloud properties in eastern China and its adjacent ocean and their relationships with meteorological conditions, ***Climatic and Environmental Research*** (in Chinese), 23(6): 737-748, 2018.
29. Li R., Ma X., Xiong F., **Jia H.**, Sha T., & Tian R.: Comparisons and evaluation of aerosol burden and optical depth in CMIP5 simulations over East Asia. ***J. Atmos. Sol.-Terr. Phys.,*** 105315, 2020.
30. Lü R., Yu X., **Jia H.**, and Xiao S.: Aerosol optical properties and direct radiative forcing at Taihu. ***Appl. Opt.,*** 56(25), 7002-7012, 2017.
31. Ma X., **Jia H.**, Yu F., and Quaas J.: Opposite aerosol index-cloud droplet effective radius correlations over major industrial regions and their adjacent oceans, ***Geophys. Res. Lett.***, 45, 5771–5778, 2018.
32. Ma X., **Jia H.**, Sha T., An J., and Tian R.: Spatial and seasonal characteristics of particulate matter and gaseous pollution in China: Implications for control policy, ***Environ. Pollut.***, 2019.
33. Ma X., and **Jia H.**: Particulate matter and gaseous pollutions in three megacities over China: Situation and implication, ***Atmos. Environ.***, 140, 476-494, 2016.
34. Ma X., Sha T., Wang J., **Jia H.**, and Tian R.: Investigating impact of emission inventories on PM2. 5 simulations over North China Plain by WRF-Chem. ***Atmos. Environ.***, 195, 125-140, 2018.
35. Ma X., Wang J., Yu F., **Jia H.**, and Hu Y.: Can MODIS AOD be employed to derive PM2. 5 in Beijing-Tianjin-Hebei over China?. ***Atmos. Res.***, 181, 250-256, 2016.
36. Sha T., Ma X., **Jia H.**, Ding J., Zhang Y., and Chang Y.: Exploring the influence of two inventories on simulated air pollutants during winter over the Yangtze River Delta. ***Atmos. Environ.***, 206, 170-182, 2019.
37. Sha T., Ma X., **Jia H.**, Tian R., Chang Y., Cao F., and Zhang Y.: Aerosol chemical component: Simulations with WRF-Chem and comparison with observations in Nanjing. ***Atmos. Environ.***, 218, 116982, 2019.
38. Tian R., Ma X., **Jia H.**, Yu F., Sha T., and Zan Y.: Aerosol radiative effects on tropospheric photochemistry with GEOS-Chem simulations. ***Atmos. Environ.,*** 208, 82-94, 2019.

Teaching/Supervising Experience

2025 – Present Supervising PhD, Chenwei Li, Space Research Organization Netherlands (SRON)
Project: *Tackling Challenges in Aerosol–Cloud InterACTIONs with a Novel Framework and New Satellites (**ACIaction**)*

2024 – Present Supervising PhD, Ardit Arifi, Space Research Organization Netherlands (SRON)
Project: *Joint Use of Aerosol and Cloud Products to Quantify Emissions and Radiative Forcing*

2024 – present Co-supervising PhD, Elise Devigne, Université de Lille
Project: *Interactions between clouds and above-cloud aerosols*

2022 – 2023 Supervising Bachelor thesis, University of Leipzig
Aerosol cloud-albedo effect observed from satellite

2022 – 2023 Teaching assistant, University of Leipzig

Course: *Thermodynamics*

Academic Service

- Reviewer for Journal: *Nature Climate Change* | *Nature Communications*
  Atmospheric Chemistry and Physics | *Geophysical Research Letters*
  npj Climate and Atmospheric Science | *npj Cliamte Action*
  Journal of Advances in Modeling Earth Systems
  Journal of Geophysical Research: Atmospheres
  Atmospheric Research | *Earth System Science Data*
  Atmospheric Environment | *Remote Sensing* | *Earth and Space Science*
  Scientific Reports | *Atmospheric Science Letters*
  Pure and Applied Geophysics | *Geoscience Letters* | *Atmosphere etc.*
- Reviewer for Funding: *Swiss National Science Foundation (SNSF)*
- Reviewer for Fellowship: *Marie Skłodowska-Curie Postdoctoral Fellowship Programme, 2025*
- ***Lead*** *of AeroCom Phase 4 Multi-Model PPE (MMPPE) experiment*
- ***Lead*** *of CMIP7 ACI-PPEMIP experiment*
- ***Board member*** *of working group "Clouds and Radiation" of the International Radiation Commission (IRC; since 2025)*
- ***Board member*** *of AeroCom Science Steering Committee*
- ***Board member*** *of HAMMOZ model Science Steering Committee*
- *Convener of the session 'Clouds, Aerosol, Radiation and Precipitation Interactions' EGU 2025&2026*
- *Convener of the session 'Current progress on Aerosol Cloud Interactions' IAC 2026*
- *Convener of the session 'Perturbed parameter ensembles' 24th AeroCom / 13th AeroSat workshop*
- *Organizer of CleanCloud Seminar Series (https://projects.au.dk/cleancloud/news/seminars)*

Conference Presentations

Talks

- 2025/12 AeroCom Forum (online) ***Invited Talk***
- 2025/10 24th AeroCom workshop/ 11th AeroSAT workshop (Paris, France)
- 2025/01 105th AMS Annual Meeting (New Orleans, USA) ***Invited Talk***
- 2024/05 EGU General Assembly (Vienna, Austria) ***Highlight Talk***
- 2022/10 21th AeroCom workshop/ 9th AeroSAT workshop (Oslo, Norway)
- 2022/09 11th International Aerosol Comference (Athens, Greece)
- 2022/05 EGU General Assembly 2022 (Vienna, Austria)
- 2022/05 Aerosol, Cloud, Precipitation and Climate (ACPC) workshop (online)
- 2021/10 20th AeroCom workshop/ 8th AeroSAT workshop (online)

Posters

- 2025/04 EGU General Assembly (Vienna, Austria)
- 2024/05 EGU General Assembly (Vienna, Austria)
- 2022/03 DA-CH Meteorology Conference (Leipzig, Germany)
- 2019/12 American Geophysical Union Fall Meeting (San Francisco, USA)
- 2019/05 International GEOS-Chem Meeting (IGC9) (Boston, USA)
- 2018/12 American Geophysical Union Fall Meeting (Washington DC, USA)
- 2018/10 AGU|CAS Joint Meeting (Shannxi, China)
- 2018/05 Regional GEOS-Chem Asia Meeting (GCA1) (Nanjing, China)